

Liquid - Liquid Extraction

Final Lab Report #2

CHE 382 - Unit Operations Lab II
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1. Executive Summary

The purpose of this experiment was to determine how different impeller speeds affect the separation efficiency of a three-stage mixer-settler unit. Some variables that were involved include agitator speed, mineral oil and acetic acid mixture composition, and water flowrate. These variables affect the overall separation observed in our system.

Agitator speeds of 400, 600, and 800 rpm were tested, respectively. Samples were taken from the three separation tanks and titrated against 0.05M standardized NaOH to determine the composition of acetic acid within the system. The concentration of acetic acid was calculated at the various impeller speeds. For example, at 400rpm the concentration of acetic acid in stages 1, 2, and 3, respectively, were 0.028, 0.014, and 0.0004 M. Our results suggest that as the impeller speed increases the concentration of acetic acid in water decreases with every stage, indicating that at higher impeller speeds the extraction occurs more efficiently. We observe that with increasing impeller speeds the Murphree efficiency increases for the extract. However, we also observe that the Murphree efficiency for the raffinate generally decreases with increasing impeller speeds in our results.

2. Introduction

Numerous processes in chemical Engineering require the separation of one or more components that forms a part of a liquid mixture by exploiting relative solubility properties in two different immiscible liquids.[1] This is usually achieved by bringing the feed mixture (in this case a mixture of mineral oils and acetic acid in solution) with a liquid solvent (water) in which the component to be extracted (acetic acid) is preferably more soluble.[2] Depending on the scenario, the purification of the liquid carrying with a certain amount of impurities determines the function for which the process is designed and in other cases, the extracting a dissolved component needed in another part of a process may be the goal. Liquid-liquid extraction, when compared to distillation, avoids the need for reboilers and condensers and as such can be energy saving. Apart from its energy saving potential, it also has the potential to separate components that are sensitive to heat while avoiding the use of other expensive alternatives such as distillation by vacuum. How effective this extraction method is depends mostly on the rate at which the dissolved component is transferred from one solvent to another, which in turn is dependent on process variables such as temperature, rate of flow and so on, as well as the area of the liquid-liquid interface. Some industrial application include oil extraction, removal of high-boiling organic components in waste-water systems and the separation of tightly bonded hydrogen bonded organic components from water (as in this experiment).[3]

In this lab experiment, a three-stage mixer-settler experimental set-up was deployed to extract acetic acid from mineral oils and water is used as the solvent. At varied agitator speeds (400, 600 and 800 rpm(s)), the effect of varying agitator speeds on the extraction efficiency and

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Also missing McCabe-Thiele result (Figure 6 in Results).

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the acetic acid composition was investigated. Doing this is important because at high agitation, the solution and solvent do not have enough residence time to separate into two separate layers, which is what's expected at lower agitator speeds. At low agitator speeds, steady state levels of liquids are somewhat difficult to achieve. [4] By finding the optimized agitator speed, solute concentration and extraction efficiency can be increased.

The Experimental set-up for this experiment consisted of a raffinate tank, a feed tank, three-stage tanks, an agitator in each stage tank, a valve that controls the water flow-rate, a valve that controls the oil flow-rate, two flow-meters for the mineral oil and water, and a pump. In the feed tank, 23 gallons of mineral oil were mixed with acetic acid such that the resulting solution contained 0.5 wt% acetic acid. Water was then ran into the mixer-settler unit until the water was at a steady state in each stage tank. Once steady state was achieved, the feed (mixture of mineral oil and acetic acid) was fed to the stage tanks at 40 gpm and the water flow-rate was adjusted to 70 gpm. The agitator speed was then set to 400 rpm and the system was run until steady state was reached. Samples of mineral oil and water were taken from each stage and their compositions were found by titrating with 0.05 M sodium hydroxide and phenolphthalein as the indicator. The agitator speeds were then changed to 600 and 800 rpm and the same procedure was repeated for each agitator speeds with the mineral oil and water flow-rates kept constant.

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3. Theory

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Liquid extraction is a counter-current separation process that transfers the solute from one solvent (the feed) to another solvent resulting in a solute rich product (the extract) and a residual liquid from which the solute has been removed (the raffinate). Usually, one of the solvents is an aqueous mixture or just water and the other solvent is a nonpolar organic liquid. For a single extraction stage, the solvent liquid and feed liquid are fed into the stage and an agitator mixes the two liquids. This mixture then aerates in the settler and separates into two liquid phases with the heavier on the bottom and the lighter on top [4]. On the other hand, in multiple stages, the extract and raffinate streams flow from one stage to the other counter-currently, producing two final raffinate and extract products. Addition of more stages results in less solvent being used and a higher degree of extraction can be achieved.

For this experiment, agitator speed in each stage was set at 400, 600, and 800 rpm(s) (in that order), to be able to qualitatively determine how the increasing speed affects the efficiency. At each agitator speed, 25 mL samples of water and mineral oil were obtained from each extraction stage compartment and titrated with NaOH to ascertain the concentration of acetic acid present judging by the amount of NaOH used to achieve the expected color change (from colorless to pink) for the phenolphthalein indicator used. Plots of acetic acid concentration in the mineral oil and water samples obtained were then made for each stage and at each speed [5]. Since the main objective is to remove the acetic acid out of the oil, the higher the acetic acid concentration in the water sample, is an indication of how efficient the extraction system is.

For the analysis of the experiment data obtained, a series of phase equilibria and mass balances for each extraction stage will be solved and from the results, the concentration of solute in stage $n+1$ as a function of the initial concentration, y_1 will be obtained as depicted in;

$$y_{n+1} = \left(\frac{E^{n+1} - 1}{E - 1} \right) y_1$$

$$\text{And } E = \frac{KL}{H}$$

Where,

E [=] a constant (the higher the value of E , the greater the extent of extraction)

K [=] Equilibrium Constant

H [=] Concentration of Feed Solvent

L [=] Concentration of Extraction solvent

Also, the efficiency, η of the system can be determined (similar to the concept of distillation) by;

$$\eta = \frac{N_A}{N_T}$$

where,

N_A [=] Actual Number of Stages

N_T [=] The Theoretical Number of Stages

With reference to the resulting plotted graphs, efficiency as related to various stages of the extraction, can be determined as follows:

At stage, S , Murphree of the extract is defined as:

$$E_{ME} = \frac{Y_s - Y_{s+1}}{Y_s - Y_{s+1}^*}$$

Where,

Y_{s+1}^* [=] composition of the extract in equilibrium with the raffinate leaving the $(S+1)^{th}$ stage.

Murphree efficiency of the raffinate is defined as:

$$E_{ME} = \frac{X_s - X_{s+1}}{X_s - X_{s+1}^*}$$

Where,

X_{s+1}^* [=] composition of the raffinate in equilibrium with the extract leaving $(S+1)^{th}$ stage

Finally, for the overall efficiency:

$$E_o = \frac{\eta_T}{\eta_{act.}}$$

Where,

η_T [=] theoretical number of stages required

η_{act} [=] actual number of stages used

For multi-component or complex systems with recycle and multiple feed, the analysis are carried out using a commercial process simulator. The programs solve the mass balances and the equilibrium relations for each stage.

4. Results

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The analysis of the concentration of acetic acid in the mixture was carried out by titrating different samples from the stages in the system with a solution of 0.05 M NaOH. In addition to this, the agitator speed was changed from 400 RPM to 600 RPM and finally to 800 RPM to see the effects of the impeller speed on the concentration of acetic acid. The trend is shown in figures 1 and 2 below.

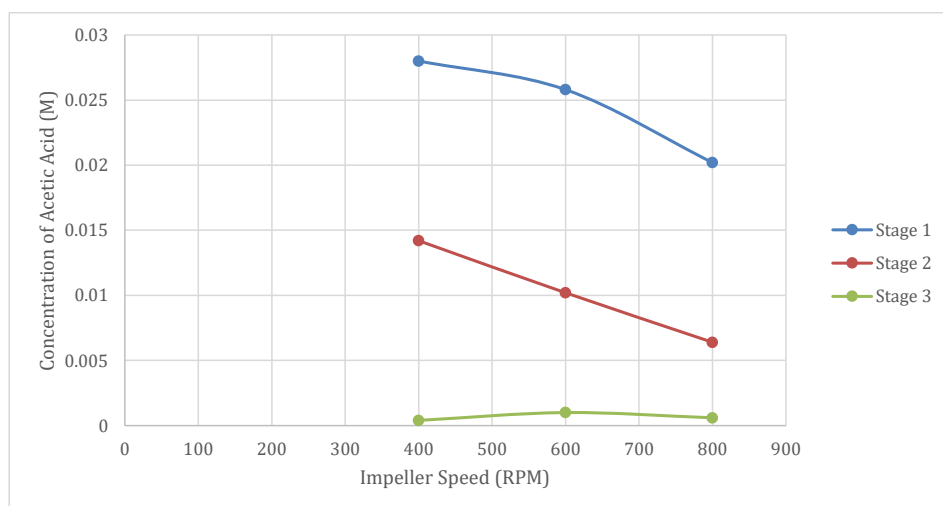


Figure 1 – Acetic Acid Concentration vs. Impeller Speed at Different Stages for Water

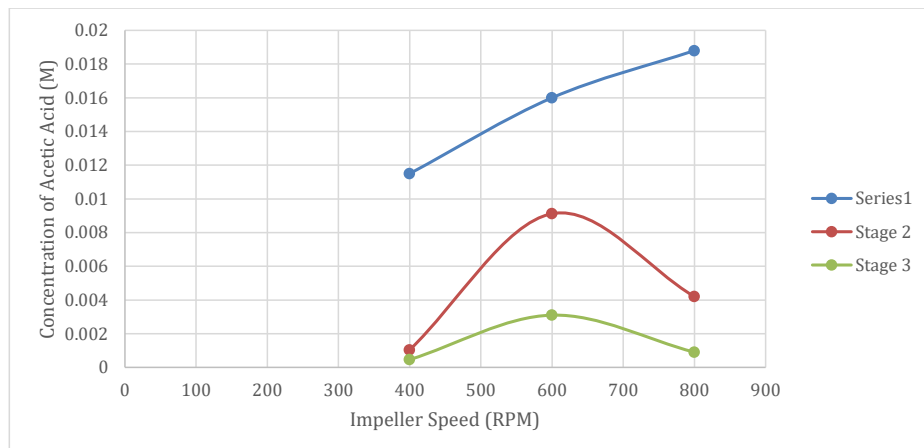


Figure 2 – Acetic Acid Concentration vs. Impeller Speed at Different Stages for White Oil

Just as shown in figure 1, as the impeller speed increases the concentration of acetic acid in water decreases per stage, indicating that at higher impeller speed, the liquid – liquid extraction method is more efficient. Nevertheless, figure 2 displays a different trend for the concentration of acetic acid in the white oil. Based on this trend, the concentration of acetic acid is relatively low at 400 RPM, goes up at 600 RPM, and finally decreases at 800 RPM for stages 2 and 3, while it keeps going up in stage 1.

A different trend may be observed when plotting the acetic acid concentration versus the number of stages at different agitator speeds. These trends are observed in figures 3 and 4 for water and oil, respectively.

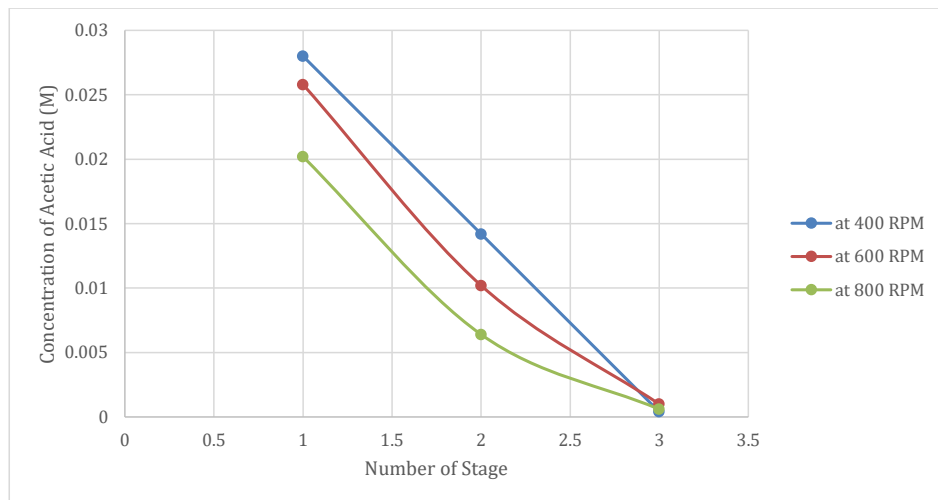


Figure 3 – Acetic Acid Concentration vs. Number of Stage at Different Impeller Speeds for Water

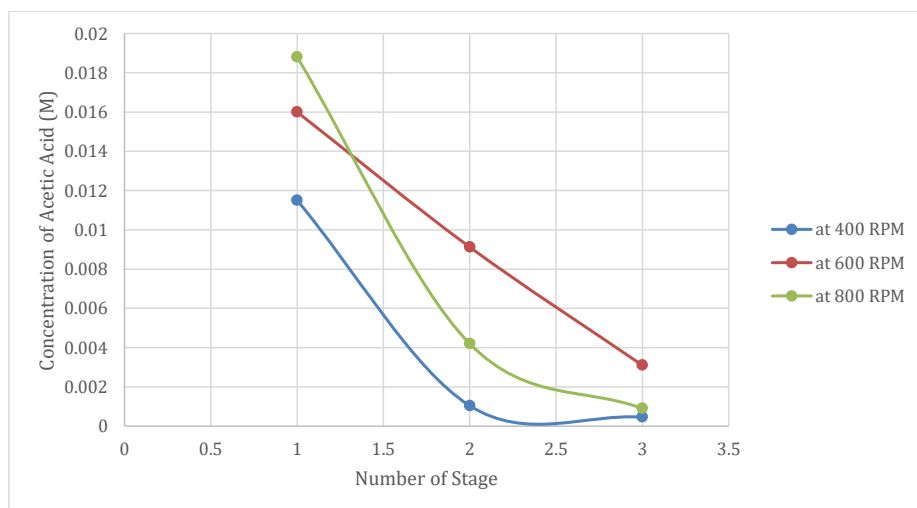


Figure 4 – Acetic Acid Concentration vs. Number of Stage at Different Impeller Speeds for White Oil

As displayed in figures 3 and 4, as the number of stage increases, the concentration of acetic acid in either the raffinate or the extract decreases as well. This trend makes sense because as the mixture is exposed to more stages, higher levels of separation are reached, and thus the extraction of acetic acid from these substances increases. Therefore, the concentration is expected to decrease.

When it comes to analyzing how the mass fraction of acetic acid changes in the raffinate and in the extract solvent, a plot of Y_A (concentration in the extract) versus X_A (concentration in the raffinate) is sketched out. This plot is displayed in figure 5 below.

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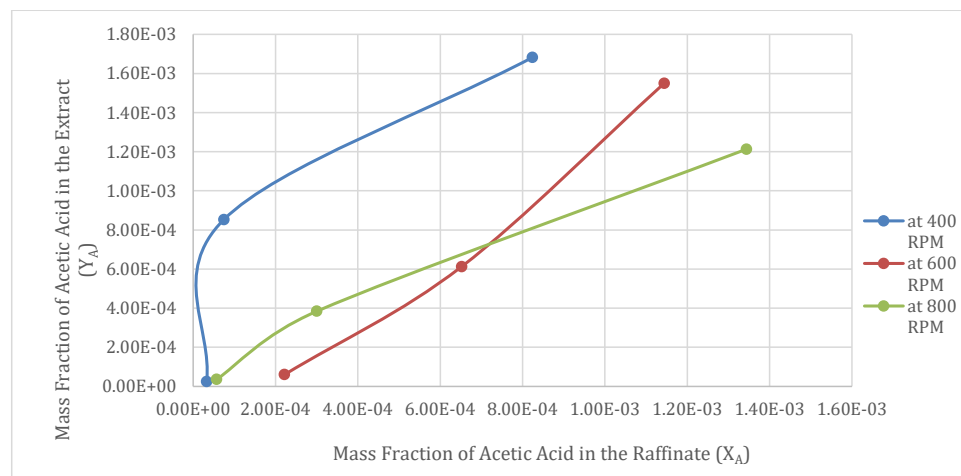


Figure 5 – Mass Fraction of Acetic Acid in Extract (Y_A) vs. Mass Fraction of Acetic Acid in Raffinate (X_A) for different impeller speeds

This figure displays how the mass fractions vary from larger to smaller ones as the impeller speed increases and the number of stage increases. Based of this, it is possible to conclude that the impeller speed makes the process more efficient for the mass fractions decrease when the impeller speed is larger.

Using the equilibrium data of the white oil, acetic acid, and water mixture we can graph it and obtain a slope of the line then use this to obtain the number of theoretical stages and proceed with the calculation of the efficiency. Figure 6 below displays figure 5 with the addition of the equilibrium data.

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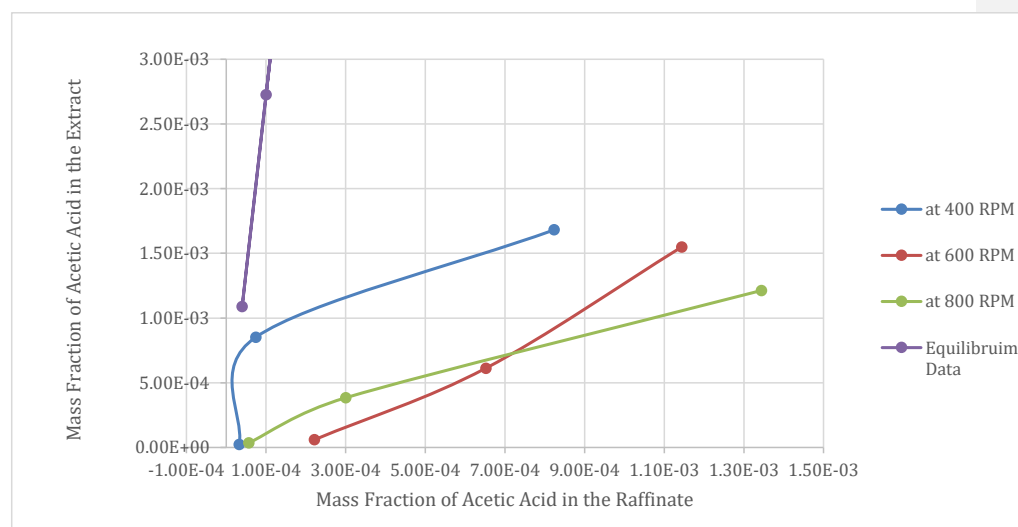


Figure 6- Mass Fraction of Acetic Acid in Extract (Y_A) vs. Mass Fraction of Acetic Acid in Raffinate (X_A) for different impeller speeds with Equilibrium Data

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Table 1 - Raffinate and Extract Murphree Stage Efficiency and Overall Efficiency

Water and Acetic Acid Murphree Efficiency(%)			
	Stage 1	Stage 2	Stage 3
400 RPM	49.2%	97.2%	100%
600 RPM	60.5%	90.2%	100%
800 RPM	68.3%	90.6%	100%
White Oil and Acetic Acid Murphree Efficiency(%)			
	Stage 1	Stage 2	Stage 3
400 RPM	90.9%	55.7%	39.2%
600 RPM	42.8%	66.0%	91.0%
800 RPM	77.6%	80.9%	65.0%
Total Efficiency = 66.7%			

5. Discussion

In chemical engineering, liquid extraction is of high interest as separation is required in most processes. Studying the effect of different variables on the efficiency and purity of product can improve the quality of the final product and make a process more economical. In this experiment, we investigated the effect of the agitator speed on the efficiencies and the acetic acid concentration in both the raffinate and the extract. When performing the extraction, no reflux was used in the extractor. It is important to also investigate how the addition of reflux would change the results.

From Table X, we can observe that as we increase the impeller speeds, we notice that the Murphree efficiency increases for the extract which consists of the water and acetic acid mixture. On the other hand, the Murphree efficiency for the raffinate including the white oil and acetic acid decreases with increasing impeller speeds in general. For the second stage, the raffinate does not follow this trend and increases its efficiency when increasing the agitator speed. However, the efficiencies calculated in table X do not reflect very accurate results as the equilibrium data given is obtained at much higher mass fractions of acetic acid than what we used in this lab. Furthermore, the efficiency requires a stage specific equilibrium mole fraction of acetic acid that we did not have. This can significantly affect our results.

When changing the agitator speed, the acetic acid concentration in the raffinate and the extract also changed. Our results indicate that with increasing impeller speeds the amount of acetic acid that is mixed with the water is not affected much at the first stage. However, it decreases the number of theoretical stages needed because the separation occurs faster. Increasing the impeller speed will also increase the rate at which the acetic acid separates from the white oil. In an industrial process, this would be an optimization problem to solve as increasing the agitator speed will increase utility cost but decrease the number of stages. An optimal point should be calculated to optimize profitability.

This experiment operated under no reflux; however, reflux extractors are very common in industry. As you increase the reflux ratio, the number of theoretical stages decreases. This is true for a continuous countercurrent extraction process which we performed. Increasing the reflux and the impeller speeds are both valid methods enhancing separation and reducing the size of an extraction column. The addition of reflux will also increase the overall efficiency of the extraction and improve the process.

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6. References

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[1] "Liquid-Liquid Extraction." *Chemistry LibreTexts*, Libretexts, 5 June 2019, chem.libretexts.org/Bookshelves/Ancillary_Materials/Demos,_Techniques,_and_Experiments/General_Lab_Techniques/Liquid-Liquid_Extraction. Accessed 19 February 2020

[2] "Liquid Liquid Extraction." *Liquid Liquid Extraction - an Overview | ScienceDirect Topics*, www.sciencedirect.com/topics/immunology-and-microbiology/liquid-liquid-extraction. Accessed 19 February 2020.

[3] "Liquid Liquid Extraction." *LIQUID LIQUID EXTRACTION*, www.stscanadainc.com/liquid-liquid-extraction.html. Accessed 19 February 2020.

[4] "Typical Industrial Applications for Separation by Extraction." Koch Modular Process Systems. <https://kochmodular.com/liquid-liquid-extraction/industrial-applications/> Accessed 19 February 2020

[5] W. McCabe, J. Smith, and P. Harriot, *Unit Operations of Chemical Engineering*, 7th ed. McGraw-Hill, 2005, pages 687 -720.

7. Appendix I: Data Tabulation/Graphs

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For the first part of the experiment, it was necessary to know the initial concentration of acetic acid in the white oil. This was achieved by collecting a 25 mL sample and titrating it with NaOH.

After knowing the concentration of acetic acid in the mixture, the data was collected and summarized in the following tables:

Table 1 – Data for the Acetic Acid Titration in Water at 400 RPM

Impeller Speed	400 RPM		
Extract	Water		
# of Stage	1	2	3
Volume of NaOH used (mL)	14.0	7.10	0.20
Moles of NaOH used	7 x 10 ⁻⁴ moles	3.55 x 10 ⁻⁴ moles	1 x 10 ⁻⁵ moles
Moles of Acetic Acid titrated			
Volume of Solution (mL)	25.0		
Concentration of Acetic Acid (M)	0.0280	0.0142	4.00 x 10 ⁻⁴
Mass fraction of Acetic Acid	1.68 x 10 ⁻³	8.53 x 10 ⁻⁴	2.40 x 10 ⁻⁵

Table 2 – Data for the Acetic Acid Titration in White Oil at 400 RPM

Impeller Speed	400 RPM		
Raffinate	White Oil		
# of Stage	1	2	3
Volume of NaOH used (mL)	5.75	0.52	0.23
Moles of NaOH used	2.88×10^{-4} moles	2.60×10^{-5} moles	1.15×10^{-5} moles
Moles of Acetic Acid			
Volume of Solution (mL)	25.0		
Concentration of Acetic Acid (M)	0.0115	1.04×10^{-3}	4.60×10^{-4}
Mass fraction of Acetic Acid	8.24×10^{-4}	7.43×10^{-5}	3.29×10^{-5}

Table 3 – Data for the Acetic Acid Titration in Water at 600 RPM

Impeller Speed	600 RPM		
Extract	Water		
# of Stage	1	2	3
Volume of NaOH used (mL)	12.9	5.10	0.50
Moles of NaOH used	6.45×10^{-4} moles	2.55×10^{-4} moles	2.50×10^{-5} moles
Moles of Acetic Acid titrated			
Volume of Solution (mL)	25.0		
Concentration of Acetic Acid (M)	0.0258	0.0102	1.00×10^{-3}
Mass fraction of Acetic Acid	1.55×10^{-3}	6.13×10^{-4}	6.01×10^{-5}

Table 4 – Data for the Acetic Acid Titration in White Oil at 600 RPM

Impeller Speed	600 RPM		
Raffinate	White Oil		
# of Stage	1	2	3
Volume of NaOH used (mL)	8.00	4.55	1.15
Moles of NaOH used	4.00×10^{-4} moles	2.28×10^{-4} moles	7.75×10^{-5} moles
Moles of Acetic Acid			
Volume of Solution (mL)	25.0		
Concentration of Acetic Acid (M)	0.0160	9.12×10^{-3}	3.10×10^{-3}
Mass fraction of Acetic Acid	1.14×10^{-3}	6.52×10^{-4}	2.22×10^{-4}

Table 5 – Data for the Acetic Acid Titration in Water at 800 RPM

Impeller Speed	800 RPM		
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Extract	Water		
# of Stage	1	2	3
Volume of NaOH used (mL)	10.1	3.20	0.30
Moles of NaOH used	5.05×10^{-4} moles	1.60×10^{-4} moles	1.5×10^{-5} moles
Moles of Acetic Acid titrated			
Volume of Solution (mL)	25.0		
Concentration of Acetic Acid (M)	0.0202	6.40×10^{-3}	6.00×10^{-4}
Mass fraction of Acetic Acid	1.21×10^{-3}	3.84×10^{-4}	3.60×10^{-5}

Table 6 – Data for the Acetic Acid Titration in White Oil at 800 RPM

Impeller Speed	800 RPM		
Raffinate	White Oil		
# of Stage	1	2	3
Volume of NaOH used (mL)	9.40	2.10	0.40
Moles of NaOH used	4.70 x 10 ⁻⁴ moles	1.05 x 10 ⁻⁴ moles	2.00 x 10 ⁻⁵ moles
Moles of Acetic Acid			
Volume of Solution (mL)	25.0		
Concentration of Acetic Acid (M)	0.0188	4.20 x 10 ⁻³	8.00 x 10 ⁻⁴
Mass fraction of Acetic Acid	1.34 x 10 ⁻³	3.00 x 10 ⁻⁴	5.72 x 10 ⁻⁵

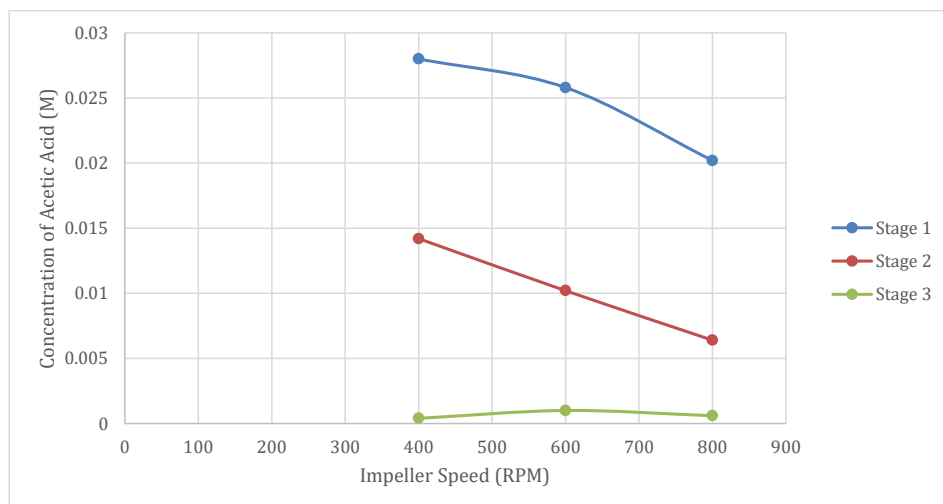


Figure 7 – Acetic Acid Concentration vs. Impeller Speed at Different Stages for Water

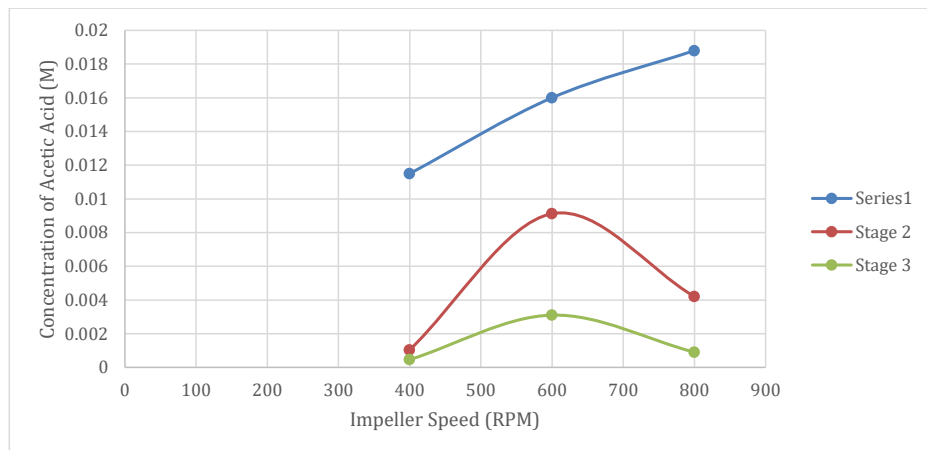


Figure 8 – Acetic Acid Concentration vs. Impeller Speed at Different Stages for White Oil

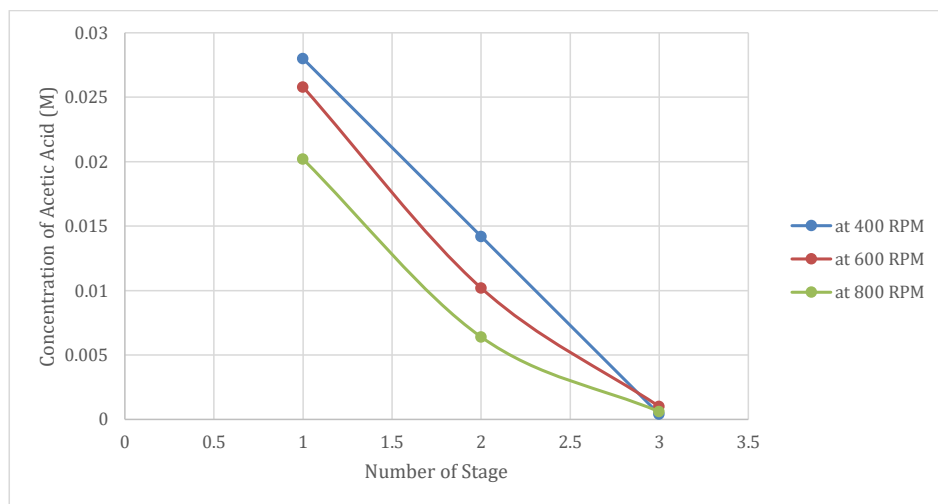


Figure 9 – Acetic Acid Concentration vs. Number of Stage at Different Impeller Speeds for Water

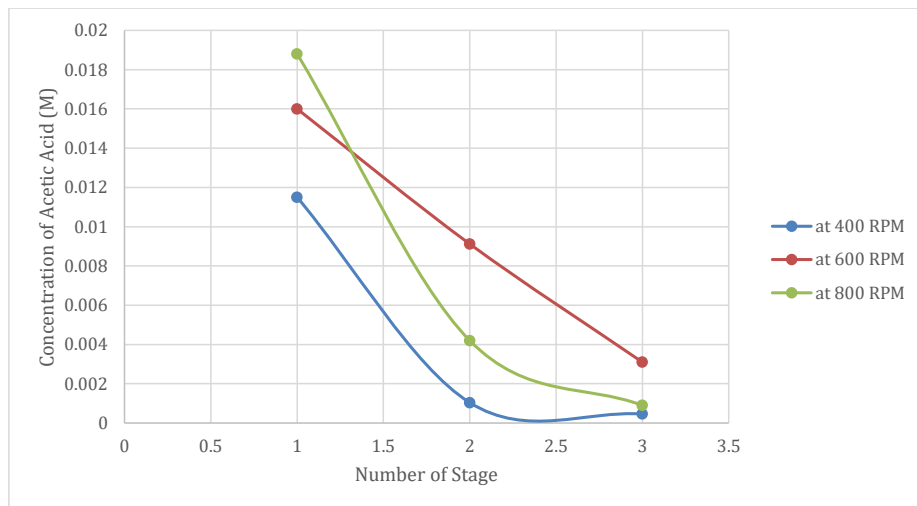


Figure 10 – Acetic Acid Concentration vs. Number of Stage at Different Impeller Speeds for White Oil

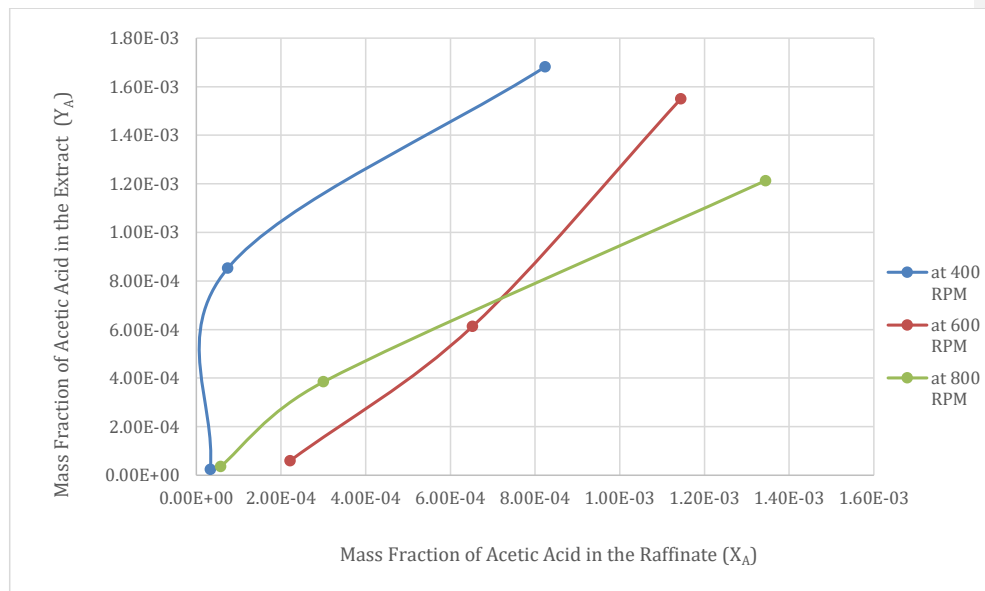
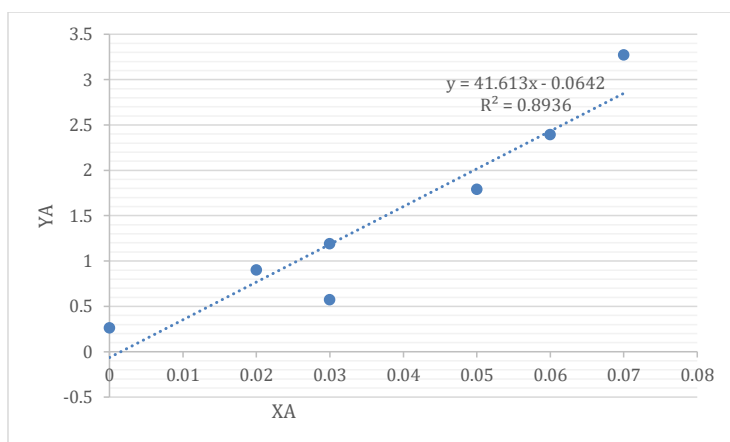


Figure 11 – Mass Fraction of Acetic Acid in Extract (Y_A) vs. Mass Fraction of Acetic Acid in Raffinate (X_A) for different impeller speeds

Table 4 – Equilibrium Data for the Water, White Oil, and Acetic Acid System

Equilibrium Data		
Initial wt% of acetic acid in oil	X_A	Y_A
5%	0.07	3.27
4%	0.06	2.39
3%	0.05	1.79
2%	0.03	1.19
1.5%	0.02	0.9
1%	0.03	0.57
0.5%	0	0.26

**Figure 12 – Equilibrium Data for the Water, White Oil, and Acetic Acid System****8. Appendix II: Error Analysis**

After performing the liquid extraction of acetic acid from a mineral water solution, we noticed many possible sources of error that may have affected the results of this experiment. The liquid extraction apparatus consisted of three separation troughs with corresponding impellers. An external handheld tachometer measured the rotation rate of the impeller. The impeller speed may have varied during the extraction without our knowledge due to the absence of a consistent tachometer measurement. Also, samples from the extraction troughs were collected both for the water and oil side of the separation. We collected the samples with a volumetric pipet to ensure the collected samples maintained a volume of 25mL. The existence of residual acetic acid in the pipet may have influenced the calculated concentration of acetic acid in the water and oil

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mixture.

To calculate the concentration of acetic acid in the collected solution, we performed a titration. The titration process requires adding an indicator with a strong base to neutralize the solution and slightly turn the solution basic. The color change of the solution is subjective to the experimenter performing the titration and is not as accurate as a pH monitor. Also, the amount of strong base added requires the measurement by the experimenter. This error can amount to a 0.1mL variance in addition to the uncertainty combined with the pH indicator. The possibility of error could be near 0.5mL with this extreme variance in volume; we expect possible significant changes in calculated acetic acid concentration.

9. Appendix III: Sample Calculation

The calculations of this experiment started off with obtaining how many mL of acetic acid had to be added to the white oil for the solution to be 0.5 wt% acetic acid. Given that the tank originally contained 23 gallons of white oil, the calculations go as follows:

$$23 \text{ gal white oil} \times \frac{3.79 \text{ L}}{1 \text{ gal}} \times \frac{10^3 \text{ mL}}{1 \text{ L}} \times \frac{0.82 \text{ g of white oil}}{1 \text{ mL}} = 71,393 \text{ g of white oil}$$

$$71,393 \text{ g of white oil} \times (0.5\% \text{ wt}) = 357 \text{ g of acetic acid}$$

$$357 \text{ g of acetic acid} \times \frac{1 \text{ mL}}{1.05 \text{ g}} = 340 \text{ mL of acetic acid}$$

Therefore, 340 mL of acetic must be used for the solution to be 0.5 wt% acetic acid.

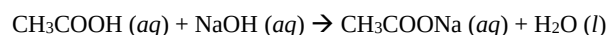
When the titration starts, a solution of 0.05 M NaOH is used. Thus, the calculations for the titration of acetic acid with NaOH in the water extract at impeller speed of 400 RPM go as follows:

Stage 1:

Volume of NaOH used = 14.0 mL

$$14.0 \text{ mL of NaOH} \times \frac{10^{-3} \text{ L}}{1 \text{ mL}} \times \frac{0.05 \text{ moles}}{1 \text{ L}} = 7 \times 10^{-4} \text{ moles of NaOH}$$

The titration of acetic with NaOH is described by the following reaction:



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Thus, the moles of NaOH used for titration are equal to the moles of acetic acid present in the solution.

$$\text{Moles of CH}_3\text{COOH} = 7 \times 10^{-4} \text{ moles of Acetic Acid}$$

$$\text{Concentration of acetic acid} = \frac{\text{moles of Acetic Acid}}{\text{L of solution}}$$

$$\text{Concentration of acetic acid} = \frac{7 \times 10^{-4} \text{ moles of Acetic Acid}}{0.025 \text{ L of solution}} = 0.028 \text{ M}$$

Stage 2:

$$\text{Volume of NaOH used} = 7.1 \text{ mL}$$

$$7.1 \text{ mL of NaOH} \times \frac{10^{-3} \text{ L}}{1 \text{ mL}} \times \frac{0.05 \text{ moles}}{1 \text{ L}} = 3.55 \times 10^{-4} \text{ moles of NaOH}$$

$$\text{Moles of CH}_3\text{COOH} = 3.55 \times 10^{-4} \text{ moles of Acetic Acid}$$

$$\text{Concentration of acetic acid} = \frac{\text{moles of Acetic Acid}}{\text{L of solution}}$$

$$\text{Concentration of acetic acid} = \frac{3.55 \times 10^{-4} \text{ moles of Acetic Acid}}{0.025 \text{ L of solution}} = 0.014 \text{ M}$$

Stage 3:

$$\text{Volume of NaOH used} = 0.2 \text{ mL}$$

$$0.2 \text{ mL of NaOH} \times \frac{10^{-3} \text{ L}}{1 \text{ mL}} \times \frac{0.05 \text{ moles}}{1 \text{ L}} = 1.00 \times 10^{-5} \text{ moles of NaOH}$$

$$\text{Moles of CH}_3\text{COOH} = 1.00 \times 10^{-5} \text{ moles of Acetic Acid}$$

$$\text{Concentration of acetic acid} = \frac{\text{moles of Acetic Acid}}{\text{L of solution}}$$

$$\text{Concentration of acetic acid} = \frac{1.00 \times 10^{-5} \text{ moles of Acetic Acid}}{0.025 \text{ L of solution}} = 4 \times 10^{-4} \text{ M}$$

The calculations for the titration of acetic acid with NaOH in white oil at impeller speed of 400 RPM go as follows:

Stage 1:

$$\text{Volume of NaOH used} = 5.75 \text{ mL}$$

$$5.75 \text{ mL of NaOH} \times \frac{10^{-3} \text{ L}}{1 \text{ mL}} \times \frac{0.05 \text{ moles}}{1 \text{ L}} = 2.88 \times 10^{-4} \text{ moles of NaOH}$$

Moles of $\text{CH}_3\text{COOH} = 2.88 \times 10^{-4} \text{ moles of Acetic Acid}$

$$\text{Concentration of acetic acid} = \frac{\text{moles of Acetic Acid}}{\text{L of solution}}$$

$$\text{Concentration of acetic acid} = \frac{2.88 \times 10^{-4} \text{ moles of Acetic Acid}}{0.025 \text{ L of solution}} = 0.012 \text{ M}$$

Stage 2:

Volume of NaOH used = 0.52 mL

$$0.52 \text{ mL of NaOH} \times \frac{10^{-3} \text{ L}}{1 \text{ mL}} \times \frac{0.05 \text{ moles}}{1 \text{ L}} = 2.60 \times 10^{-5} \text{ moles of NaOH}$$

Moles of $\text{CH}_3\text{COOH} = 2.60 \times 10^{-5} \text{ moles of Acetic Acid}$

$$\text{Concentration of acetic acid} = \frac{\text{moles of Acetic Acid}}{\text{L of solution}}$$

$$\text{Concentration of acetic acid} = \frac{2.60 \times 10^{-5} \text{ moles of Acetic Acid}}{0.025 \text{ L of solution}} = 1.04 \times 10^{-3} \text{ M}$$

Stage 3:

Volume of NaOH used = 0.23 mL

$$0.23 \text{ mL of NaOH} \times \frac{10^{-3} \text{ L}}{1 \text{ mL}} \times \frac{0.05 \text{ moles}}{1 \text{ L}} = 1.15 \times 10^{-5} \text{ moles of NaOH}$$

Moles of $\text{CH}_3\text{COOH} = 1.15 \times 10^{-5} \text{ moles of Acetic Acid}$

$$\text{Concentration of acetic acid} = \frac{\text{moles of Acetic Acid}}{\text{L of solution}}$$

$$\text{Concentration of acetic acid} = \frac{1.15 \times 10^{-5} \text{ moles of Acetic Acid}}{0.025 \text{ L of solution}} = 4.6 \times 10^{-4} \text{ M}$$

Similarly, the other calculations for different impeller speeds were performed.

Next, the calculation for the mass fractions in the raffinate (X_A) and in extract (Y_A) have to be carried out. These fractions are defined as follows:

$$X_A = \frac{\text{mass of Acetic Acid}}{(\text{mass of Acetic Acid} + \text{mass of white oil})}$$

$$Y_A = \frac{\text{mass of Acetic Acid}}{(\text{mass of Acetic Acid} + \text{mass of water})}$$

Given that the concentration of acetic acid is diluted in water and oil respectively, the mass of solution will be obtained through their respective densities.

$$25 \text{ mL of solution} \times \frac{1 \text{ g}}{1 \text{ mL}} = 25 \text{ g of solution (water)}$$

$$25 \text{ mL of solution} \times \frac{0.84 \text{ g}}{1 \text{ mL}} = 21 \text{ g of solution (oil)}$$

Just like the sample calculations above, these are for the impeller speed of 400 RPM. The calculations at different speeds follow the same pattern.

Stage 1:

$$X_A = \frac{(2.88 \times 10^{-4} \text{ moles}) \left(\frac{60.05 \text{ g}}{1 \text{ mol}} \right)}{21 \text{ g of solution}} = 8.24 \times 10^{-4}$$

$$Y_A = \frac{(7 \times 10^{-4} \text{ moles}) \left(\frac{60.05 \text{ g}}{1 \text{ mol}} \right)}{25 \text{ g of solution}} = 1.68 \times 10^{-3}$$

Stage 2:

$$X_A = \frac{(2.60 \times 10^{-4} \text{ moles}) \left(\frac{60.05 \text{ g}}{1 \text{ mol}} \right)}{21 \text{ g of solution}} = 7.43 \times 10^{-4}$$

$$Y_A = \frac{(3.55 \times 10^{-4} \text{ moles}) \left(\frac{60.05 \text{ g}}{1 \text{ mol}} \right)}{25 \text{ g of solution}} = 8.53 \times 10^{-4}$$

Stage 3:

$$X_A = \frac{(1.15 \times 10^{-5} \text{ moles}) \left(\frac{60.05 \text{ g}}{1 \text{ mol}} \right)}{21 \text{ g of solution}} = 3.29 \times 10^{-5}$$

$$Y_A = \frac{(1 \times 10^{-5} \text{ moles}) \left(\frac{60.05 \text{ g}}{1 \text{ mol}} \right)}{25 \text{ g of solution}} = 2.40 \times 10^{-5}$$

Using the equilibrium data of the white oil, acetic acid, and water mixture we can graph it and obtain a slope of the line then use this to obtain the number of theoretical stages and proceed with the calculation of the efficiency. Since the equilibrium data shown in figure 6 does not match the data in lab we cannot perform the McCabe Thiele method. Furthermore, the ternary diagram of white oil, acetic acid and water is not provided, rather only the kerosene, acetic acid and water mixture data is shown. A reasonable approximation of the required number of stages to reach ideal separation is 2 obtained when kerosene equilibrium data is used. This number will be used to obtain the Murphree and overall efficiency.

Murphree efficiency:

Raffinate=White Oil

$$E_{MR} = \frac{X_s - X_{s+1}}{X_s - X_{s+1}^*}$$

For stage 1 at 400 RPM for acetic acid in white oil,

$$E_{MR} = \frac{X_1 - X_2}{X_1 - X_2^*}$$

X_1	X_2	X_2^*
1.68×10^{-3}	8.53×10^{-4}	9×10^{-4}

$$E_{MR} = \frac{1.68 \times 10^{-3} - 8.53 \times 10^{-4}}{1.68 \times 10^{-3} - 9 \times 10^{-4}}$$

$$E_{MR} = 0.613$$

Extract=Water

$$E_{ME} = \frac{Y_{s+1} - Y_s}{Y_{s+1}^* - Y_s}$$

For stage 1 at 400 RPM for acetic acid in water extract,

$$E_{ME} = \frac{Y_2 - Y_1}{Y_2^* - Y_1}$$

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Y_1	Y_2	Y_2^*
8.24×10^{-5}	7.43×10^{-4}	8.13×10^{-4}

$$E_{ME} = \frac{7.43 \times 10^{-4} - 8.24 \times 10^{-5}}{8.13 \times 10^{-4} - 8.24 \times 10^{-5}}$$

$$E_{ME} = 0.904$$

Overall Efficiency:

$$E_o = \frac{\eta_T}{\eta_{act}}$$

$$E_o = \frac{2}{3}$$

$$E_o = 0.67$$

10. Individual Team Contributions

Name: Javier Eduardo Mendieta	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	4	Record data/ Measured
Pre-Lab Editing	2	Contributed
Experimental Objective (Prelab)	1	Contributed
Apparatus (prelab)	1	Contributed
Materials & Supplies (prelab)	1	Contributed
Procedure (prelab)	2	Contributed
Project Safety Evaluation (prelab)	2	Contributed

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Final Lab Editing	2	Contributed
Executive Summary (final lab)	2	Contributed
Introduction (Final lab)	3	Contributed
Theory (Final Lab)	3	Contributed
Results (final lab)	2	Contributed
Discussion (final lab)	2	Contributed
Conclusion (final lab)	2	Contributed
References (final and/or prelab)	1	Contributed
Data Tabulation/Graphs (final)	2	Completed
Error Analysis (final lab)	1	Contributed
Sample Calculations (final Lab)	3	Completed
PowerPoint Presentation	1	Contributed
Total	37	

Name: Omar Zahdan	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	2	Record data/ Measured
Pre-Lab Editing	2	Contributed

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Experimental Objective (Prelab)	2	Contributed
Apparatus (prelab)	2	Contributed
Materials & Supplies (prelab)	2	Contributed
Procedure (prelab)	2	Completed
Project Safety Evaluation (prelab)	1	Contributed
Final Lab Editing	3	Contributed
Executive Summary (final lab)	2	Contributed
Introduction (Final lab)	2	Contributed
Theory (Final Lab)	1	Contributed
Results (final lab)	2	Contributed
Discussion (final lab)	2	Contributed
Conclusion (final lab)	2	Contributed
References (final and/or prelab)	1	Contributed
Data Tabulation/Graphs (final)	1	Contributed
Error Analysis (final lab)	2	Completed
Sample Calculations (final Lab)	2	Contributed
PowerPoint Presentation	1	Contributed

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Total	34	
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Name: Omar Takrouri	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	2	Record data/ Measured
Pre-Lab Editing	3	Contributed
Experimental Objective (Prelab)	1	Contributed
Apparatus (prelab)	2	Contributed
Materials & Supplies (prelab)	1	Contributed
Procedure (prelab)	1	Contributed
Project Safety Evaluation (prelab)	1	Contributed
Final Lab Editing	1	Contributed
Executive Summary (final lab)	1	Contributed
Introduction (Final lab)	2	Completed
Theory (Final Lab)	1	Contributed
Results (final lab)	2	Contributed
Discussion (final lab)	2	Completed
Conclusion (final lab)	2	Contributed

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References (final and/or prelab)	3	Contributed
Data Tabulation/Graphs (final)	2	Contributed
Error Analysis (final lab)	1	Contributed
Sample Calculations (final Lab)	1	Contributed
PowerPoint Presentation	2	Contributed
Total	31	

Name: Joel Finlon	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	2	Record data/ Measured
Pre-Lab Editing	1	Contributed
Experimental Objective (Prelab)	1	Contributed
Apparatus (prelab)	1	Contributed
Materials & Supplies (prelab)	1	Contributed
Procedure (prelab)	2	Contributed
Project Safety Evaluation (prelab)	1	Contributed
Final Lab Editing	4	Contributed
Executive Summary (final lab)	2	Contributed

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Introduction (Final lab)	2	Contributed
Theory (Final Lab)	2	Completed
Results (final lab)	1	Contributed
Discussion (final lab)	1	Contributed
Conclusion (final lab)	2	Completed
References (final and/or prelab)	3	Contributed
Data Tabulation/Graphs (final)	1	Contributed
Error Analysis (final lab)	2	Contributed
Sample Calculations (final Lab)	1	Contributed
PowerPoint Presentation	1	Contributed
Total	31	

Name: Yusuf Sorunke	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	2	Record data/ Measured
Pre-Lab Editing	1	Contributed
Experimental Objective (Prelab)	1	Contributed
Apparatus (prelab)	1	Contributed

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Materials & Supplies (prelab)	1	Contributed
Procedure (prelab)	2	Contributed
Project Safety Evaluation (prelab)	1	Contributed
Final Lab Editing	4	Contributed
Executive Summary (final lab)	2	Contributed
Introduction (Final lab)	2	Contributed
Theory (Final Lab)	2	Completed
Results (final lab)	1	Contributed
Discussion (final lab)	1	Contributed
Conclusion (final lab)	2	Completed
References (final and/or prelab)	2	Contributed
Data Tabulation/Graphs (final)	1	Contributed
Error Analysis (final lab)	2	Contributed
Sample Calculations (final Lab)	1	Contributed
PowerPoint Presentation	1	Contributed
Total	30	

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*CHE 382 - Unit Operations Lab II Group #4
Thursday, PM Section*

Spring 2020 02/19/2020